6. $p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$

$\vec{x}_n$ - выборка

a) ОМП

$$L(\theta, \vec{x}_n) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{(\theta-1)^n}{\left(\prod_{i=1}^{n} x_i\right)^\theta} \{x_{min} \geq 1\}$$

$$\ln L(\theta) = n \ln(\theta - 1) - \theta \left(\sum_{i=1}^{n} \ln x_i\right)$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta - 1} - \sum_{i=1}^{n} \ln x_i = 0 \Rightarrow \theta = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i}$$

$$\frac{\partial^2 \ln L}{\partial \theta^2} = \frac{-n}{(\theta-1)^2} < 0 \Rightarrow$$ локальный максимум

$$\tilde{\theta} = 1 + \frac{1}{\overline{\ln x}}$$

c) $I(\theta) = M\left[\left(\frac{\partial \ln g}{\partial \theta}\right)^2\right]$

$$\ln g = \ln(\theta - 1) - \theta \ln x$$

$$\left(\frac{\partial \ln g}{\partial \theta}\right)^2 = \left(\frac{1}{\theta - 1} - \ln x\right)^2$$

$$I(\theta) = \int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - 2\frac{\ln x}{\theta - 1} + \ln^2 x\right) \frac{\theta - 1}{x^\theta} dx =$$

$$= \int_1^{+\infty} \left(\frac{x^{-\theta}}{\theta - 1} - 2\frac{\ln x}{x^\theta} + (\theta - 1)\frac{\ln^2 x}{x^\theta}\right) dx =$$

$$= \frac{1}{(\theta-1)^2} - 2\int_1^{+\infty} \ln x \, d\left(\frac{-1}{(\theta-1)x^{\theta-1}}\right) + \int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^\theta} dx =$$

$$= \frac{1}{(\theta-1)^2} - 2\left[\ln x \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} - \int_1^{+\infty} \frac{x^{-\theta}}{1-\theta}dx\right] +$$

$$+ \int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^2}dx = \frac{-1}{(1-\theta)^2} - \int_1^{+\infty} \ln^2 x \, d(x^{1-\theta}) =$$

$$= \frac{-1}{(1-\theta)^2} - \left[\ln^2 x \cdot x^{1-\theta}\Big|_1^{+\infty} - 2\int_1^{+\infty} \frac{\ln x}{x} x^{1-\theta}dx\right] =$$

$$= \frac{-1}{(1-\theta)^2} + 2\int_1^{+\infty} \ln x \, d\left(\frac{x^{1-\theta}}{1-\theta}\right) = \frac{-1}{(1-\theta)^2} +$$

$$+ 2\left[\frac{x^{1-\theta}\ln x}{1-\theta}\Big|_1^{+\infty} - \int_1^{+\infty} \frac{x^{-\theta}}{1-\theta}dx\right] = \frac{-1}{(1-\theta)^2} + 2\frac{1}{(1-\theta)^2} = \frac{1}{(1-\theta)^2}$$

$$\frac{\sqrt{n}(\tilde{\theta}-\theta)}{\tilde{\theta}-1} \rightsquigarrow N(0,1)$$

$$t_1 < \sqrt{n}\ \overline{\ln x}\left(1 + \frac{1}{\overline{\ln x}} - \theta\right) < t_2$$

$$t_1 < \sqrt{n}\left((1-\theta)\overline{\ln x} + 1\right) < t_2$$

$$\frac{t_1}{\sqrt{n}} - 1 < (1-\theta)\overline{\ln x} < \frac{t_2}{\sqrt{n}} - 1$$

$$\frac{t_1 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < 1 - \theta < \frac{t_2 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}}$$

$$1 - \frac{t_2 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < \theta < 1 - \frac{t_1 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}}$$

в) $\frac{\partial p}{\partial \theta} = \frac{(1-\theta \ln x + \ln x)}{x^{\theta}}$

$$\int_1^{+\infty} \frac{\partial p}{\partial \theta} dx = \int_1^{+\infty} \left[(1-\theta)x^{-\theta} \ln x + x^{-\theta}\right] dx =$$

$$= \frac{x^{-\theta+1}}{1-\theta}\Big|_1^{+\infty} + \int_1^{+\infty} \ln x \, d\left(x^{1-\theta}\right) = \frac{1}{\theta-1} - \int_1^{+\infty} x^{-\theta} dx = 0$$

$$\int_1^{+\infty} \frac{\partial^2 p}{\partial x^2} dx = \int_1^{+\infty} \left(\theta x^{-\theta} \ln^2 x - x^{-\theta} \ln^2 x - 2x^{-\theta} \ln x\right) dx =$$

$$= \int_1^{+\infty} \left((\theta-1)x^{-\theta} \ln^2 x - 2x^{-\theta} \ln x\right) dx = 0$$

$$\int_1^{x_2} \frac{\theta-1}{x^{\theta}} dx = \frac{-1}{\hat{x}^{\theta-1}} + 1 = \frac{1}{2}$$

$$\hat{x} = 2^{\frac{1}{\theta-1}}$$

$$g(\tilde{\theta}) = \sqrt[\tilde{\theta}-1]{2}$$

$$\sqrt{n} \frac{g(\tilde{\theta}) - g(\theta)}{\sqrt{\nabla^T g(\tilde{\theta}) I^{-1}(\tilde{\theta}) \nabla g(\tilde{\theta})}} \rightsquigarrow N(0,1)$$

$I(\tilde{\theta})$ - ИСТР на $(1, +\infty)$ (пункт с))

$$\nabla g(\tilde{\theta}) = 2^{1/\tilde{\theta}-1} \cdot \ln 2 \cdot \frac{-1}{(\tilde{\theta}-1)^2}$$

$$\sqrt{n} \frac{(g(\tilde{\theta}) - g(\theta))(\tilde{\theta}-1)}{2^{1/\tilde{\theta}-1} \cdot \ln 2} \rightsquigarrow N(0,1)$$

$$g(\tilde{\theta}) - \frac{1{,}96 \cdot 2^{1/\tilde{\theta}-1} \ln 2}{\sqrt{n}(\tilde{\theta}-1)} < \hat{x} < g(\tilde{\theta}) + \frac{1{,}96 \cdot 2^{\frac{1}{\tilde{\theta}-1}} \ln 2}{\sqrt{n}(\tilde{\theta}-1)}$$